

1. (10 pts) Analytical Fourier Transforms

a. Compute the zero-frequency component of a single-exponential function.

$$FT \left[e^{-(\alpha|t|)} \right]_{\omega=0} =$$

(1) @ $\omega=0$
 $FT = \int_{-\infty}^{\infty} f(t) dt$

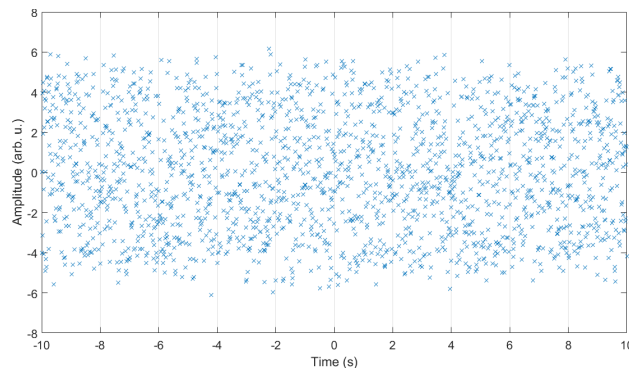
(2) So: $\int_{-\infty}^{\infty} e^{-\alpha|t|} dt =$

(3) Since (2) is even: $(2) = 2 \int_0^{\infty} e^{-\alpha t} dt$
 $FT = \frac{2}{\alpha}$

b. What is the zero-frequency component of any arbitrary function?

The zero-frequency component of any function is simply the total area under the function

2. (30 pts) Numerical Fourier transforms The following data series is mostly random noise but has a periodic function “hidden” underneath the noise. The noise is, in fact, ten times higher than the periodic signal. You can download the data set as a text file from canvas (CH368_ProblemSet2_PeriodicFunction.txt). The file contains the time axis in the first column and the data points in the second column, values are separated by tabs.



a. Carry out a numerical Fourier transformation of the data and find the frequency of the

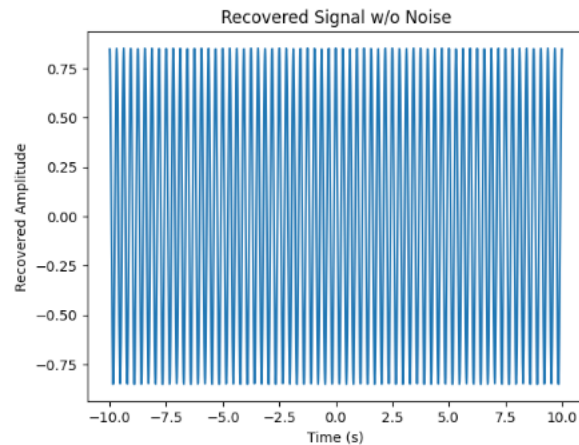
signal?

The frequency of this signal is ~ 3.2 Hz

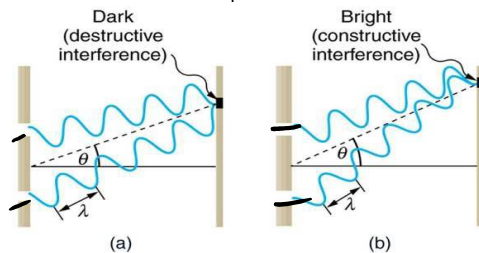
- b. Explain how you are able to recover a signal even though the noise level is much higher than the actual signal.

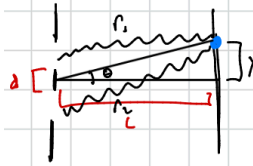
Since the noise among the signal is spread across multiple frequencies, there should be a dominant signal that is concentrated at one frequency. The FFT separates the signal from the noise

- c. Make a plot of the recovered signal as a function of time, without noise.



3. (10 pts) The double slit. Consider the double-slit experiment as shown below. If the two slits are separated by a distance d , the screen is placed at a distance L , and the wavelength is λ . Derive an expression to compute the distance between the bright and dark spots at the screen. Write down every step in your derivation, and clearly state any assumptions that you made along the way. Note: This is a simply geometry problem, it helps to draw cartoons and indicate all the variables in the cartoon as in the example below.





Geometry only

$$\Delta r = r_2 - r_1$$

$$\Delta r \approx d \sin \theta$$

$$\tan \theta = \frac{y}{L}$$

Since θ is small, $\theta = \frac{y}{L}$ * Small angle approx; assuming $L \gg d$

1) Thus $\Delta r = d \frac{y}{L}$

Bright parts

Since waves are in phase;

$$\Delta r = m\lambda \quad \text{where } m \in \mathbb{Z}, m \geq 0$$

By (1); $\frac{dy}{L} = m\lambda$

$$y_{\text{bright}} = \frac{m\lambda L}{d}$$

Dark parts

Waves out of phase by $\frac{1}{2}\lambda$

$$\text{Thus } \Delta r = \lambda \left(m + \frac{1}{2}\right) \quad \text{where } m \in \mathbb{Z}, m \geq 0$$

By (1); $\frac{dy}{L} = \lambda \left(m + \frac{1}{2}\right)$

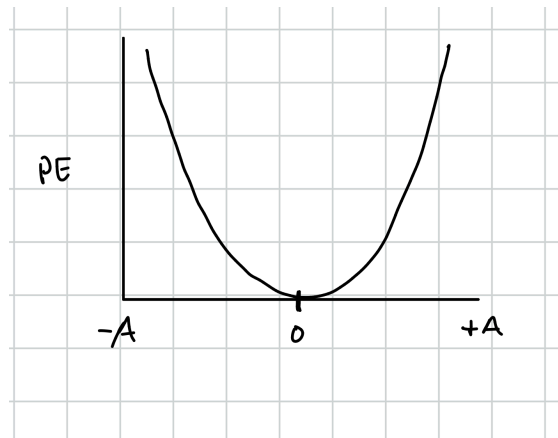
$$y_{\text{dark}} = \frac{\lambda L}{d} \left(m + \frac{1}{2}\right)$$

Distance between

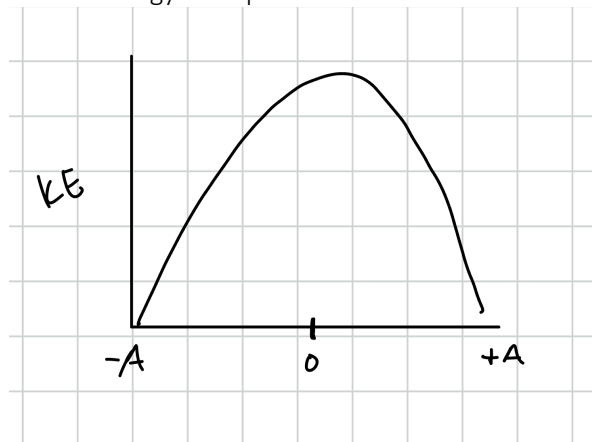
$$y_{\text{dark}} - y_{\text{bright}} = \frac{\lambda L}{d} \left(m + \frac{1}{2}\right) - \frac{m\lambda L}{d}$$

$$\Delta y_{\text{bright-dark}} = \frac{\lambda L}{2d}$$

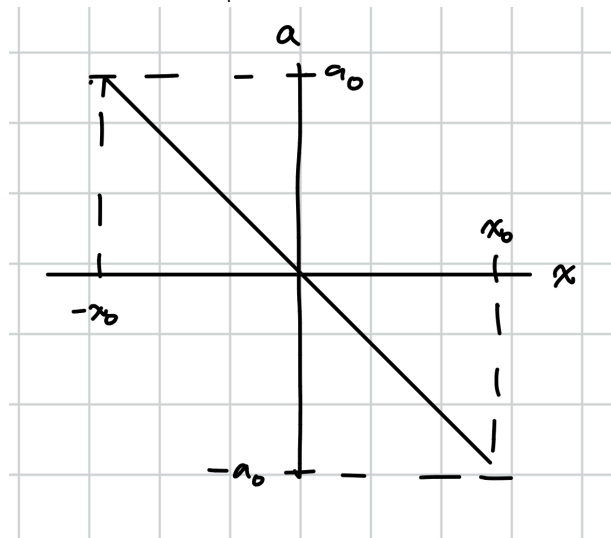
4. (10 pts) Harmonic Oscillator Consider the simple harmonic oscillator model as we discussed in lecture. Make the following graphs:
- Potential energy vs displacement



b. Kinetic energy vs displacement



c. Acceleration vs displacement



5. (20 pts) Hofmeister Series Consider this scenario: You are a scientist wishing to investigate the Hofmeister series of ions. Your task is to investigate the effects of ions on biomolecule solubility and stability. You choose to use a hypothetical protein (ProteinX), and a hypothetical DNA as the systems of interest.

a. Describe what experiments can be carried out to investigate how cations and anions affect the stability of the protein as well as the DNA. Describe specific measurements and potential sample conditions (salt concentrations etc).

For the protein, stability can be tested using differential scanning fluorimetry (DSF) to measure the melting temperatures in buffers with different salts. Several cations and anions should be used in these buffers at different concentrations to test the thermal stability.

For DNA, you could test the UV melting at 260 nm to measure the melting temperatures in the same ion series.

- b. Provide a prediction of the outcome of the proposed experiments.

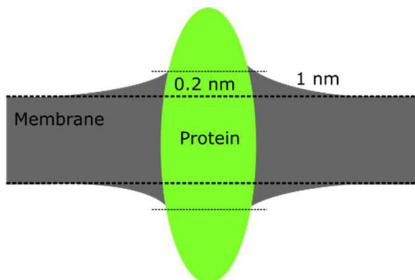
For ProteinX, adding moderate salt will often increase stability at first by shielding charged groups on the protein surface. At very high salt, the protein may become less soluble and precipitate.

For DNA, increasing salt should usually increase stability because the positive ions shield the negative phosphate groups on the backbone. Mg^{2+} and Ca^{2+} should stabilize DNA more strongly than Na^{+} or K^{+} .

- c. Describe what the differences may be between a Protein and DNA. Explain why you may observe differences in terms of the molecular electrostatics and other interactions with ions or with the solvent.

The biggest difference is that DNA is negatively charged along its backbone, so ions mainly stabilize it by screening charge repulsion. Proteins may have different charges throughout the entire surface, so ions can affect the electrostatic interactions and solubility. This is why DNA usually becomes more stable with added salt more predictably versus the unpredictable behavior of proteins in different salts.

6. (20 pts) Membrane Deformation Consider a protein embedded in a lipid membrane. The protein deforms the lipid membrane by increasing its thickness around it. If the increase in thickness is 0.2 nm. The thickness returns exponentially towards equilibrium away from the protein with a constant of 1 nm^{-1} . What is the energy associated with this deformation given that the membrane thickness force constant K is $60 \text{ k}_B\text{T}/\text{nm}^2$ ($k_B\text{T}$ is the thermal energy at 298 kelvin).



$$(1) \quad h(x) = h_0 e^{-Ax}$$

$$\text{where } h = 0.2 \text{ nm}$$

$$A = 1 \text{ nm}^{-1}$$

Elastic energy is given by;

$$(2) \quad E = \frac{1}{2} k \int_0^\infty h(x)^2 dx$$

$$\text{Deformation given by } h(x)^2 = h_0^2 e^{-2Ax}$$

$$\text{Combining (2); } E = k \int_0^\infty h_0^2 e^{-2Ax} dx$$

$$\text{Evaluating; } E = kh_0^2 \cdot \frac{1}{2A}$$

Plug in;

$$E \approx 1.2 k_B T$$